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Reasoning Fine-Tuning Induces Persistent Latent Policy States

Abir Harrasse¹, Michael Lan¹, Hunar Batra², Fateme Hashemi Chaleshtori³, Chaithanya Bandi¹

¹Martian ²University of Oxford ³University of Utah

TL;DR

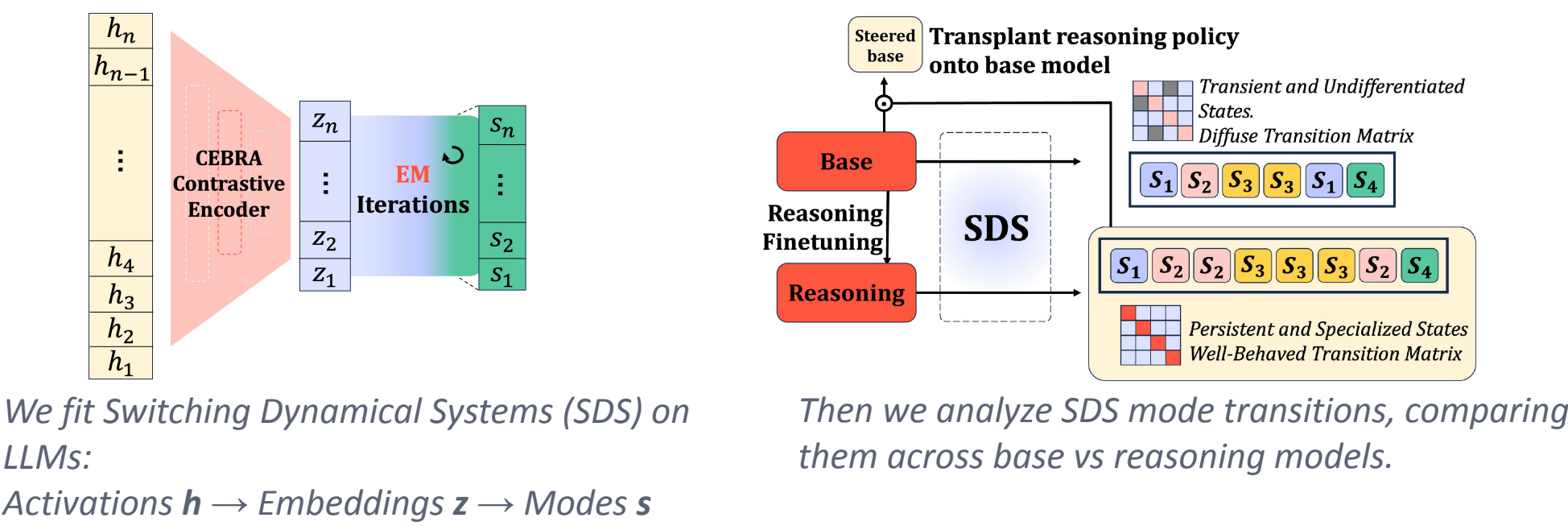
We find evidence that reasoning fine-tuning reorganizes a model’s internal dynamics into a few persistent, stage-specific latent modes. We infer these modes, propose why reasoning fine-tuning reorganizes model internals into these stages, and use the modes to both 1) steer base models and 2) prune bad reasoning steps.

1 What changes inside a fine-tuned reasoner?

Reasoning fine-tuning gives big gains on multi-step problems, but questions linger on how it alters model internals. Does it *globally reorganize* how the model internally structures its reasoning?

Main hypothesis

Reasoning is governed by a few persistent latent policy states (modes) that shape internal dynamics over many steps, and fine-tuning reorganizes them. We test this by comparing each base model with its reasoning-fine-tuned counterpart.



2 Our lens: reasoning as switching between modes

We model a reasoning trace as a system that switches between a few hidden modes, each with its own dynamics. One step = one sentence of a CoT trace (represented by its last token’s residual-stream activation; results here use the last layer). To infer modes (or “regimes”), we:

1 Embed

A contrastive encoder (CEBRA) maps adjacent steps of a trace close together and distant steps far apart.

2 Fit

Switching dynamical system (SDS): each mode has its own linear rule for moving the embedding. Fit by EM: guess modes, refit rules, repeat.

3 Decode

Read off the most likely mode at each step: which mode, how long it stays, when it switches.

$$P(s_{t+1} = j \mid s_t = i) = T_{ij}, \quad z_{t+1} = A_{s_t} z_t + b_{s_t} + \varepsilon_t, \quad \varepsilon_t \mid s_t \sim \mathcal{N}(0, \Sigma_{s_t})$$

Left: A Markov Chain over hidden modes. Modes switch with probabilities T_{ij} (that mode i is followed by mode j).

Right: Current mode s_t has its own linear rule (A, b) and noise level Σ to give next sentence’s embedding z_{t+1} from z_t

3 What we measure

TVD · total variation distance

How far switching is from random (higher = more structured)

K eff · effective number of modes

How many modes the model actually visits

p stay · stickiness

Fraction of steps that stay in the current mode

Spectral gap · memory

How fast a mode is forgotten (lower = longer memory)

MODELS. Llama-3.1-8B, Qwen2.5-14B, Qwen2.5-Math-1.5B vs. their DeepSeek-R1-distilled versions; and Qwen2.5-32B vs. QwQ-32B.

BENCHMARKS. GSM8K, SVAMP, MATH-500 (math); MMLU-Pro (multi-domain).

4 Why would fine-tuning make modes persistent?

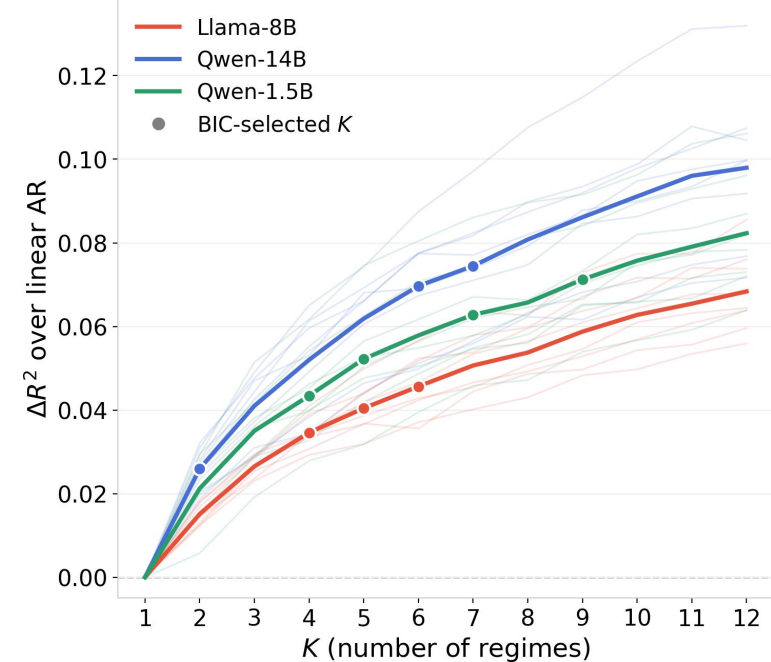
Hypothesis: Trajectory-level rewards favor sticking with a strategy

Under reinforcement learning with verifiable rewards (RLVR), a whole trace is rewarded for a checkable answer. This may favor coherent strategies: when continuing the current strategy pays more than switching, modes become sticky.

Modes may transfer via distillation

Most of the reasoning models we test are not directly trained via RLVR; they are trained to imitate traces from an RLVR-trained teacher (DeepSeek-R1). Thus, models may inherit these reasoning strategies via distillation.

5 Evidence that modes are not an artifact of clustering

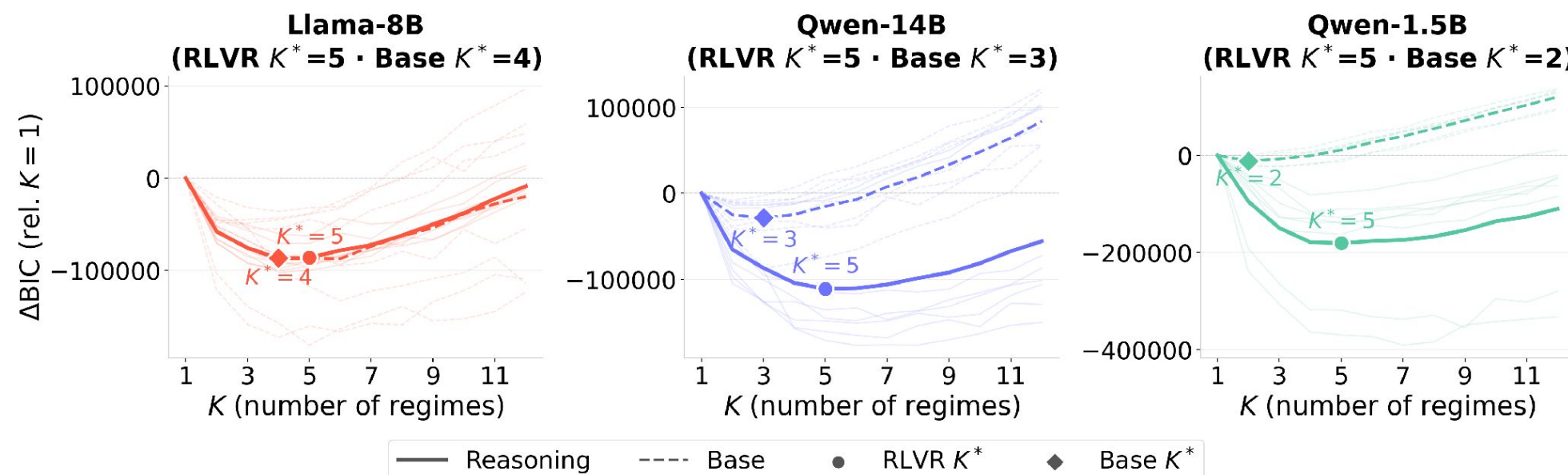


SDS (K>1) VS linear model (K=1)

AIM: Predict each sentence-level embedding from the previous one. A single linear dynamical model serves as the baseline VS SDS’s predictions.

RESULT: Allowing different latent states to have different transition rules improves prediction. The improvement tapers off after a small number of states.

The Bayesian information criterion (BIC) trades-off “fit vs model size” (lower BIC is better). Here, it selects a small K.



6 Reasoning fine-tuning reorganizes the modes

	TVD $\uparrow$		K eff $\uparrow$		p stay $\uparrow$		Spec. gap $\downarrow$	
Model	Base	Reas.	Base	Reas.	Base	Reas.	Base	Reas.
Llama-8B	0.48	0.60	4.60	4.65	0.70	0.82	0.20	0.10
Qwen-1.5B	0.39	0.61	2.60	5.30	0.78	0.79	0.24	0.07
Qwen-14B	0.43	0.58	3.00	5.80	0.78	0.77	0.17	0.12
Qwen2.5 / QwQ-32B	0.55	0.60	5.25	5.59	0.74	0.79	0.14	0.11

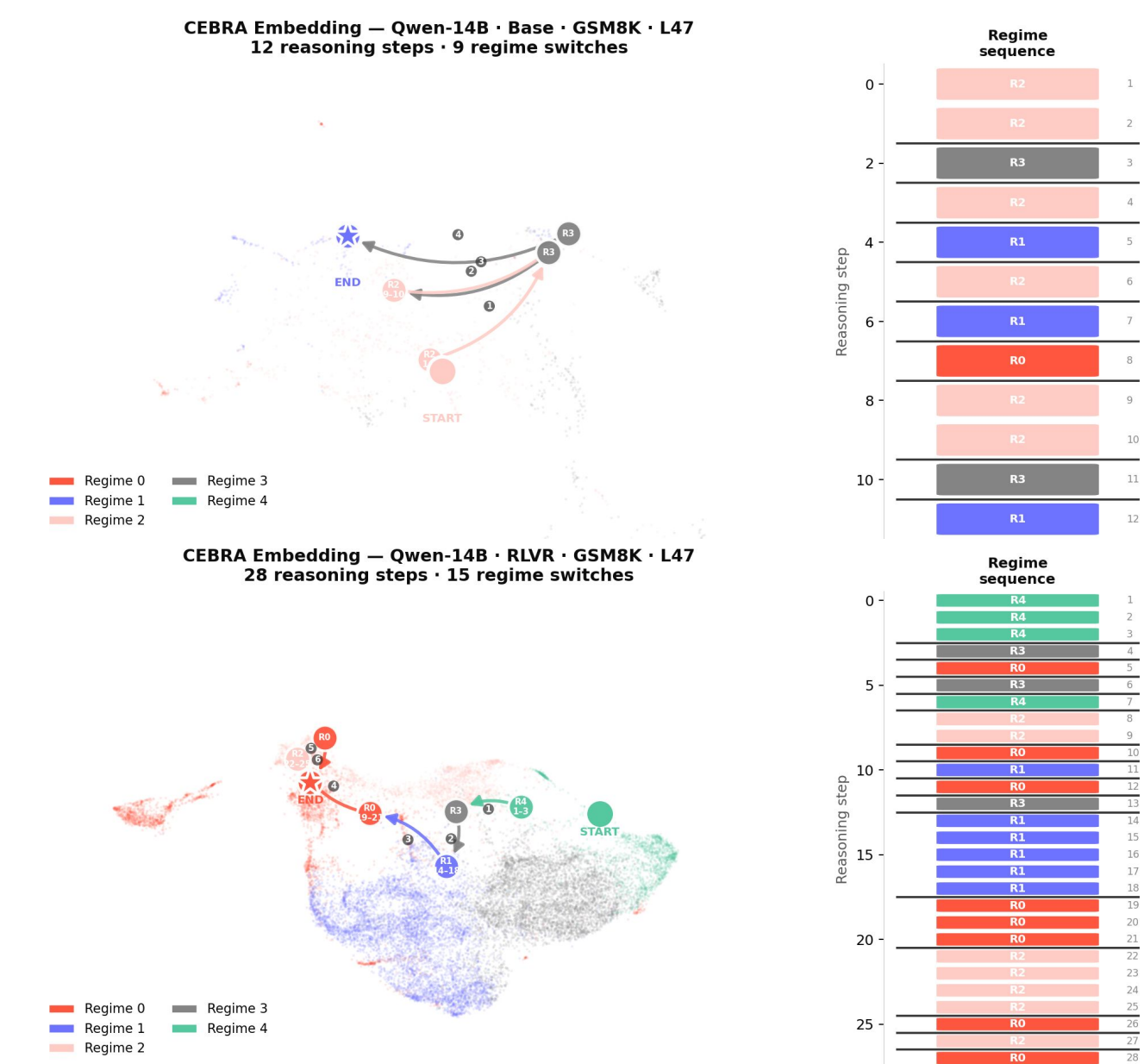
Each value is averaged over the four benchmarks and random seeds

Switching becomes more structured in every family (TVD $\uparrow$). Other metrics move in model-specific ways: Qwen models use far more modes, Llama-8B holds each mode longer, QwQ-32B improves on all four.

	TVD $\uparrow$	GSM8K		MATH-500		SVAMP		MMLU-Pro	
Model		Base	Reas.	Base	Reas.	Base	Reas.	Base	Reas.
Llama-8B	0.51	0.57	0.41	0.57	0.47	0.61	0.53	0.64	
Qwen-1.5B	0.42	0.61	0.38	0.57	0.37	0.61	0.39	0.64	
Qwen-14B	0.52	0.62	0.39	0.42	0.37	0.64	0.45	0.66	
Qwen2.5 / QwQ-32B	0.54	0.59	0.50	0.56	0.55	0.62	0.60	0.65	

TVD per benchmark: higher for the reasoning model in all model–benchmark settings (means over seeds)

The base model wanders, the reasoning model marches



Dots = sentence embeddings colored by mode; arrows trace one solution from START to END; the strip shows the mode at each step. Base: short, disordered segments. Reasoning: long segments through separated clusters.

7 Modes line up with human reasoning stages

A separate LLM judge (Qwen2.5-7B-Instruct) labels every sentence with a reasoning stage (rows). Each row shows how that stage’s sentences split across the modes R0–R4 (columns) and sums to 1. For Qwen-1.5B on GSM8K:

Reasoning Stage	R0	R1	R2	R3	R4
SETUP	0.08	0.22	0.18	0.25	0.27
FACTS	0.06	0.34	0.16	0.21	0.24
PLAN	0.08	0.10	0.18	0.31	0.33
UNCERT	0.00	0.78	0.05	0.01	0.15
CHECK	0.00	0.68	0.15	0.04	0.13
CONSOL	0.35	0.28	0.11	0.10	0.15
COMPUTE	0.00	0.38	0.15	0.27	0.20
EMIT	0.02	0.35	0.17	0.37	0.09

A self-monitoring mode

One mode (R1) carries 78% of uncertainty and 68% of checking sentences.

A separate answer mode

Emitting the final answer concentrates in a mode of its own (R3 here) in all three models.

Diffuse setup and planning

Transitional stages spread across low-specificity modes rather than owning one.

8 The modes are actionable

Transplant the reasoning model’s dynamics into its base counterpart

While the base model generates, nudge its residual stream toward the transitions the reasoning model would make. On problems the base model never solves (pass@8 = 0, i.e. none of 8 sampled attempts is correct):

0.60

pass@8 after transplant
Qwen-1.5B · GSM8K

0.46

pass@8 after transplant
Llama-8B · GSM8K

0.33

pass@8 after transplant
Llama-8B · MATH-500

PrefixGuard: prune prefixes that wander into failure modes

Can we catch a bad solution early? Each mode gets a value from how often traces passing through it end correct; “detour” modes show up mostly in traces that end incorrectly. Let a trace’s detour score be the share of its steps spent in detour modes. Start with 4 solutions but write only the first quarter of each. Keep the one that spends the least time in detour modes. Finish that trace and check if its answer is correct, then compute accuracy over each dataset.

Model	Dataset	Self-Cons.	PrefixGuard	Gain	Oracle
Llama-8B	GSM8K	60.9	68.8	+7.9	94.3
	MATH-500	18.8	25.5	+6.7	54.7
	MMLU-Pro	22.4	26.0	+3.6	70.3
	SVAMP	54.2	58.3	+4.1	94.4
Qwen-14B	GSM8K	87.5	91.7	+4.2	91.7
	MATH-500	5.5	10.2	+4.7	21.1
	MMLU-Pro	30.2	28.1	−2.1	66.7
	SVAMP	87.5	100.0	+12.5	100.0
Qwen-1.5B	GSM8K	20.3	20.8	+0.5	44.3
	MATH-500	13.5	16.7	+3.2	34.9
	MMLU-Pro	8.9	9.9	+1.0	41.1
	SVAMP	68.8	76.6	+7.8	89.1

Accuracy (%) of PrefixGuard (prune via SDS) vs. self-consistency (sample full traces, majority-vote the answer).

Oracle (best case ceiling): the accuracy if answer were always chosen perfectly from the candidate traces

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settings where PrefixGuard
beats self-consistency

+4.5

mean gain (points)

+12.5

largest gain (Qwen-14B,
SVAMP)

9 Takeaways

- We find evidence that reasoning fine-tuning globally reorganizes latent dynamics: models hold and switch between coherent computational modes over long spans.
- Trajectory-level rewards may help stabilize coherent reasoning strategies (policies); distillation may transfer this organization into smaller models.
- These models can be practically actionable. For instance, latent modes can serve as a unit for monitoring and steering reasoning behaviour.